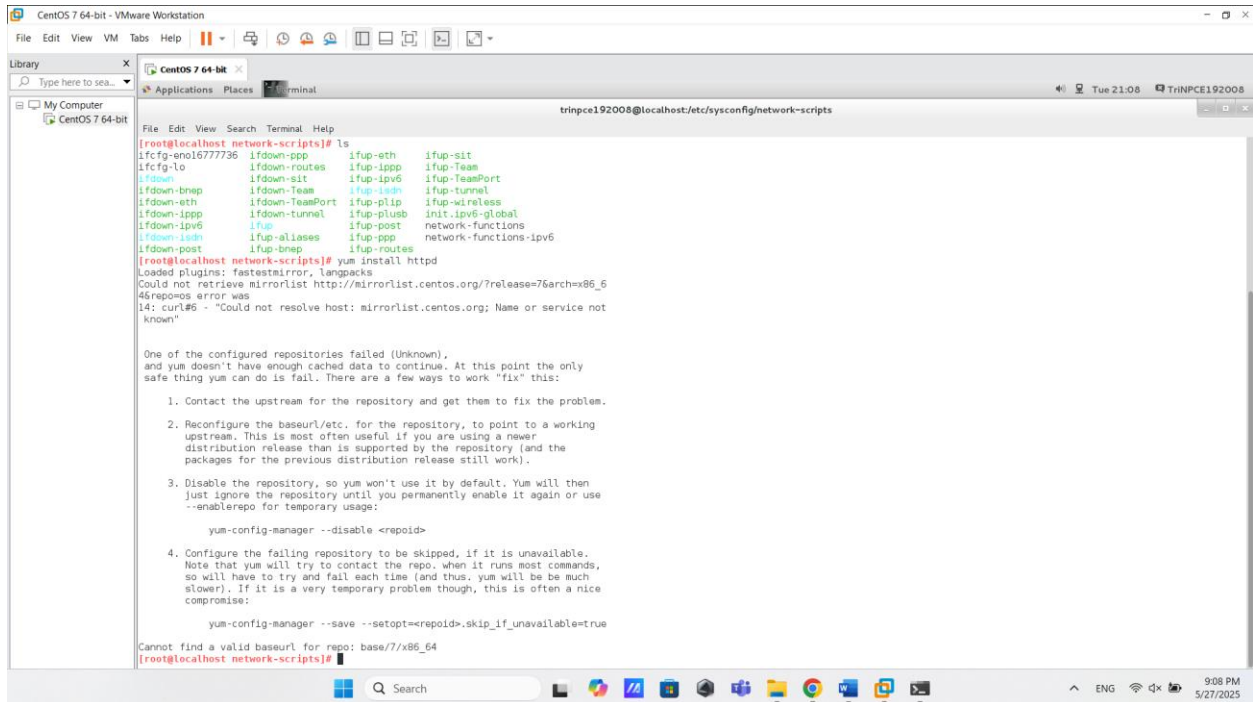


YUM:

1. Yum từ internet



The screenshot shows a terminal window in a VMware Workstation environment. The terminal is running a CentOS 7 64-bit VM. The user is in the directory `/etc/sysconfig/network-scripts` and has run the command `ls`, which lists various network scripts like `ifcfg-eno16777736`, `ifdown-ppp`, `ifup-eth`, etc. Then, the user runs `yum install httpd`. The terminal output shows that yum failed to retrieve the mirrorlist from `http://mirrorlist.centos.org/?release=7&arch=x86_64&repo=os` because it could not resolve the host. The error message suggests several ways to fix this, including contacting the upstream, reconfiguring the baseurl, disabling the repository, or configuring the failing repository to be skipped.

```
[root@localhost network-scripts]# ls
ifcfg-eno16777736  ifdown-ppp      ifup-eth        ifup-sit
ifcfg-lo          ifdown-routes  ifup-ppp        ifup-team
ifdown-eth        ifdown-sit     ifup-ipv6       ifup-teamport
ifdown-bnep       ifdown-team    ifup-iscn       ifup-tunnel
ifdown-eth        ifdown-teamport ifup-plip       ifup-wireless
ifdown-ppp        ifdown-tunnel  ifup-plusb      init.ipv6-global
ifdown-ipv6       ifup          ifup-post       network-functions
ifdown-iscn       ifup-aliases  ifup-ppp        network-functions-ipv6
ifdown-post       ifup-bnep      ifup-routes

[root@localhost network-scripts]# yum install httpd
Loaded plugins: fastestmirror, langpacks
Could not retrieve mirrorlist http://mirrorlist.centos.org/?release=7&arch=x86_64&repo=os error was
[4: curl#6 - "Could not resolve host: mirrorlist.centos.org; Name or service not known"]

One of the configured repositories failed (Unknown),
and yum doesn't have enough cached data to continue. At this point the only
safe thing yum can do is fail. There are a few ways to work "fix" this:

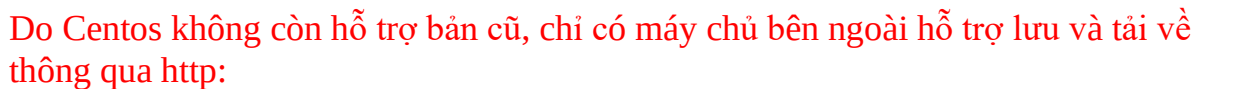
1. Contact the upstream for the repository and get them to fix the problem.
2. Reconfigure the baseurl/etc. for the repository, to point to a working
   upstream. This is most often useful if you are using a newer
   distribution release than is supported by the repository (and the
   packages for the previous distribution release still work).
3. Disable the repository, so yum won't use it by default. Yum will then
   just ignore the repository until you permanently enable it again or use
   --enablerepo for temporary usage:

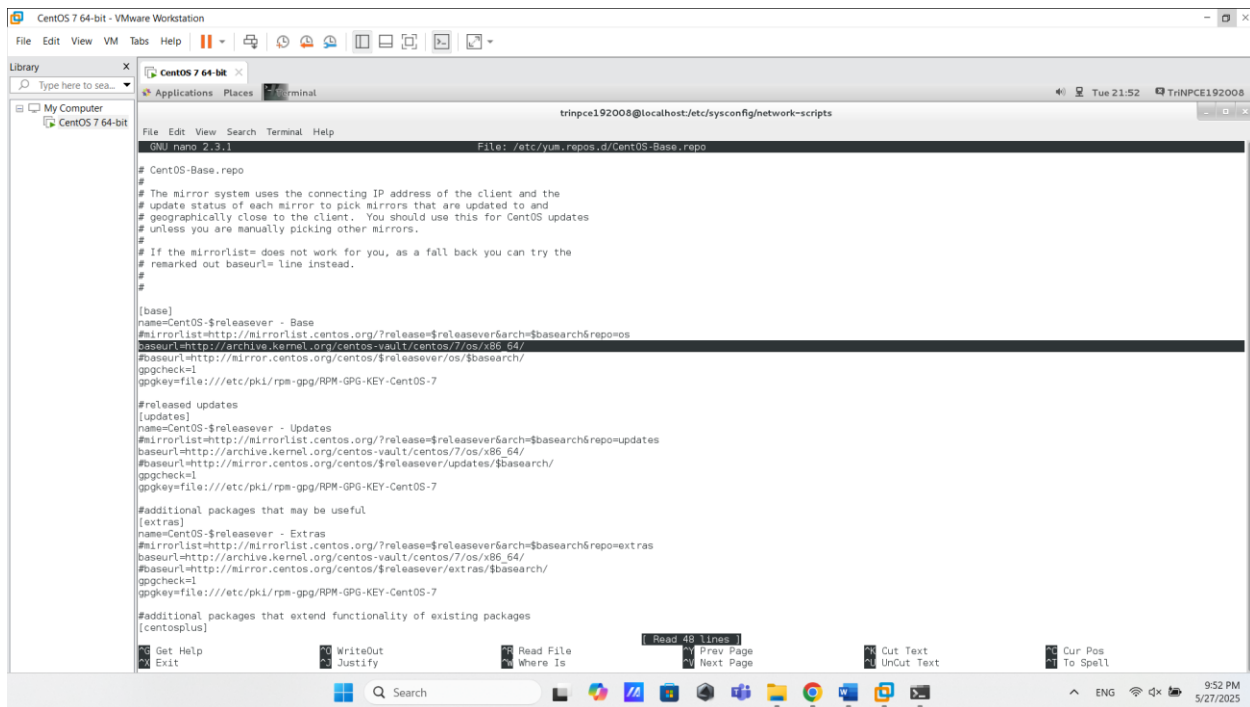
    yum-config-manager --disable <repo>
4. Configure the failing repository to be skipped, if it is unavailable.
   Note that yum will try to contact the repo, when it runs most commands,
   so will have to try and fail each time (and thus, yum will be be much
   slower). If it is a very temporary problem though, this is often a nice
   compromise:

    yum-config-manager --save --setopt=<repo>.skip_if_unavailable=true

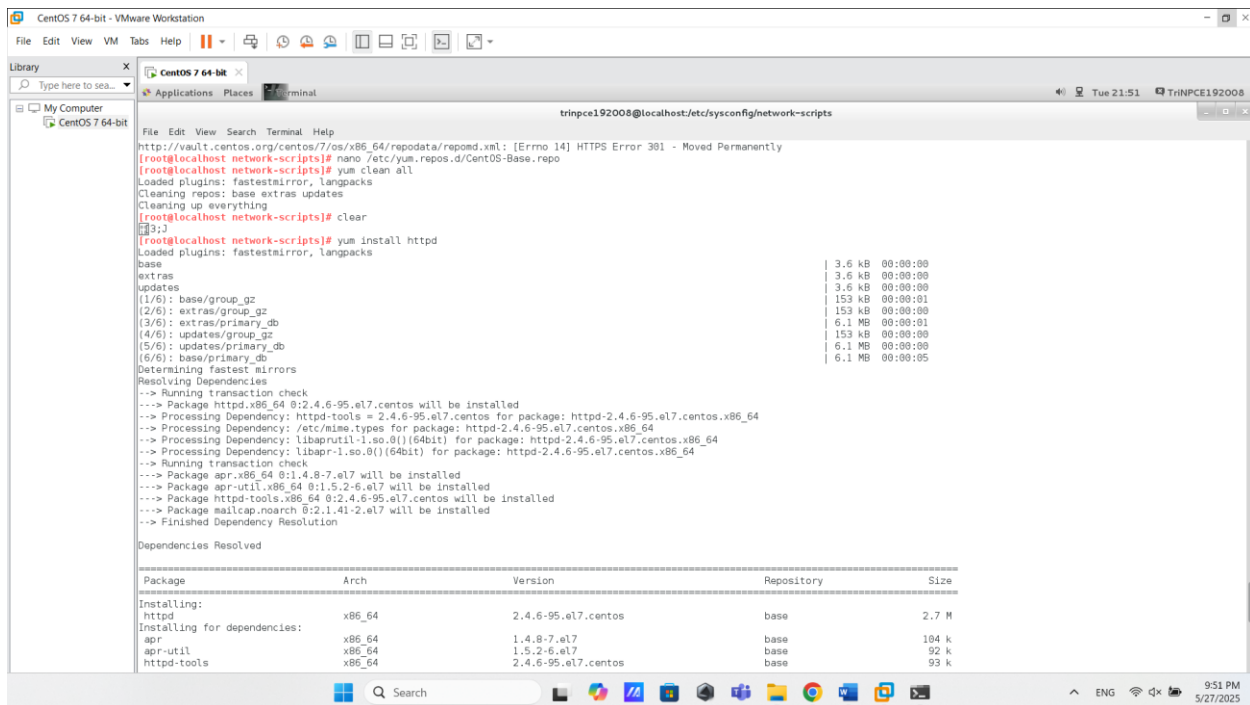
Cannot find a valid baseurl for repo: base7/x86_64
[root@localhost network-scripts]#
```

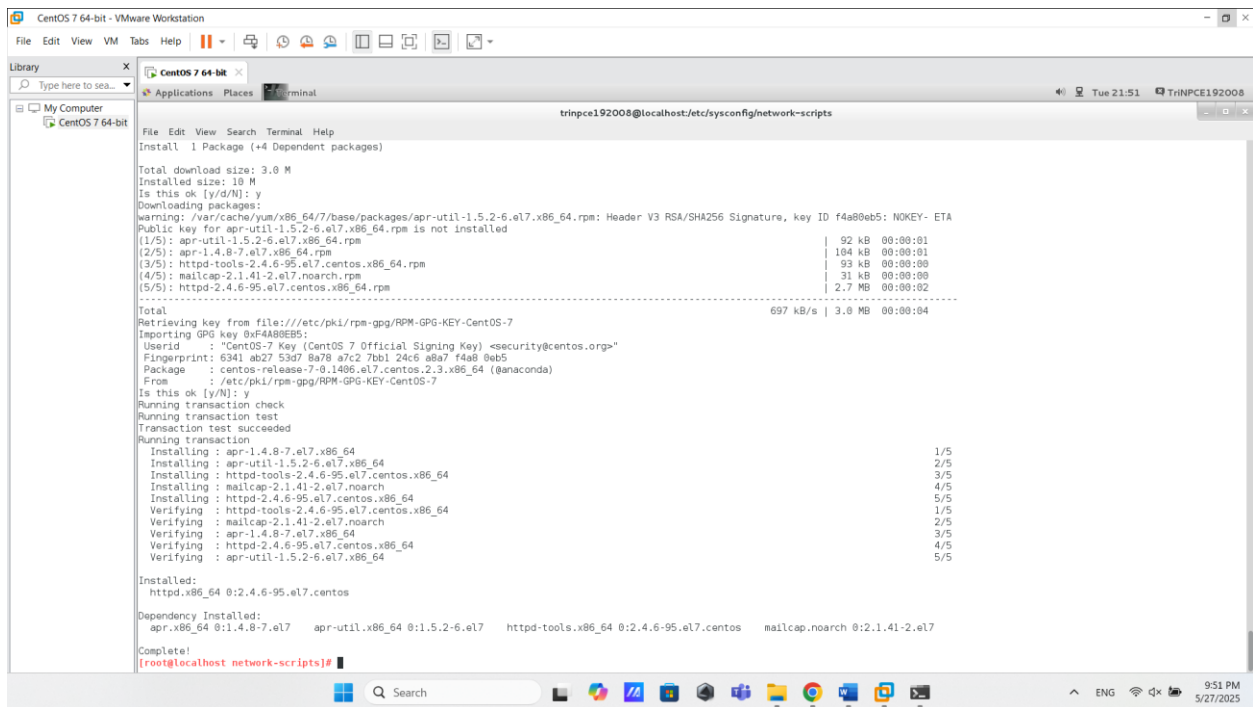
- yum cố lấy danh sách gói phần mềm từ máy chủ `mirrorlist.centos.org`.
- Nhưng vì CentOS 7 đã chuyển kho chính thức về `vault.centos.org`,
- => yum không thể tải danh sách gói => lỗi
- sửa trong file `/etc/yum.repos.d/CentOS-Base.repo`
-
- `#mirrorlist=http://mirrorlist.centos.org/?release=$releasever&arch=$basearch&repo=os&infra=$infra`
- và
- `baseurl=http://vault.centos.org/centos/$releasever/os/$basearch/`





ảnh trên: Sửa link thành http://archive.kernel.org/centos-vault/centos/7/os/x86_64/

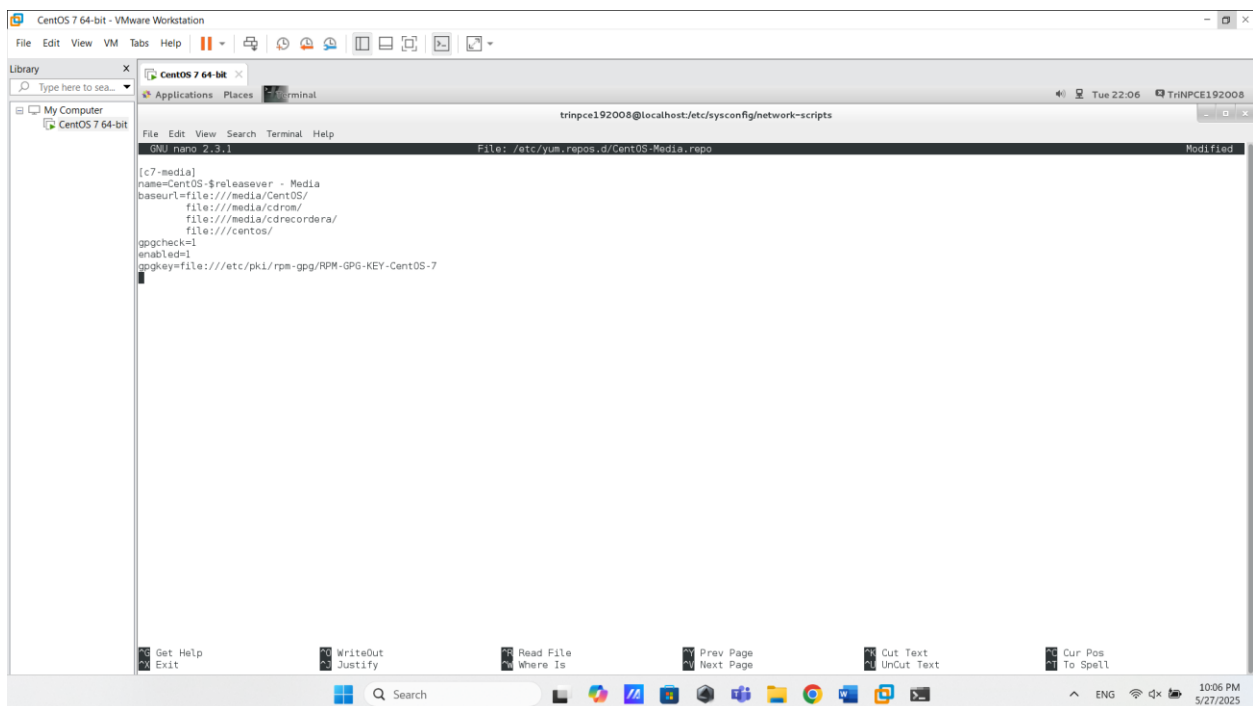




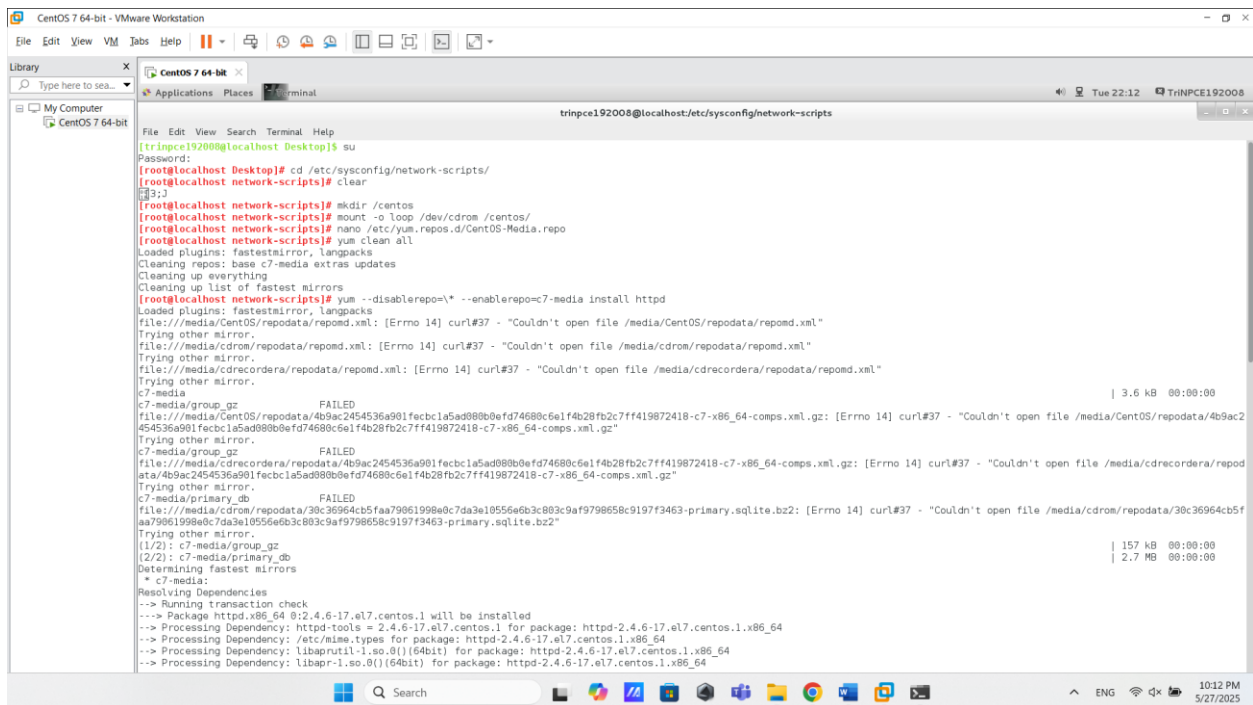
```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
CentOS 7 64-bit
trinpce192008@localhost/etc/sysconfig/network-scripts
Install 1 Package (+4 Dependent packages)
Total download size: 3.0 M
Installed size: 10 M
Is this ok [y/d/N]: y
Downloading packages:
warning: /var/cache/yum/x86_64/7/base/packages/apr-util-1.5.2-6.el7.x86_64.rpm: Header V3 RSA/SHA256 Signature, key ID f4a80eb5: NOKEY- ETA
Public key for apr-util-1.5.2-6.el7.x86_64.rpm is not installed
(1/5): apr-util-1.5.2-6.el7.x86_64.rpm | 92 kB 00:00:01
(2/5): apr-1.4.8-7.el7.x86_64.rpm | 104 kB 00:00:01
(3/5): httpd-tools-2.4.6-95.el7.centos.x86_64.rpm | 93 kB 00:00:00
(4/5): mailcap-2.1.41-2.el7.noarch.rpm | 31 kB 00:00:00
(5/5): httpd-2.4.6-95.el7.centos.x86_64.rpm | 2.7 MB 00:00:02
-----
Total | 697 kB/s | 3.0 MB 00:00:04
Retrieving key from file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7
Importing GPG key 0xF4A80EB5:
Userid : "CentOS-7 Key (CentOS 7 Official Signing Key) <security@centos.org>"
Fingerprint: 6341 ab27 53d7 8a78 a7c2 7bb1 24c6 a8a7 f4a8 beb5
Package : centos-release-7-0.1406.el7.centos.2.3.x86_64 (@anaconda)
From : /etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7
Is this ok [y/N]: y
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Installing : apr-1.4.8-7.el7.x86_64 1/5
Installing : apr-util-1.5.2-6.el7.x86_64 2/5
Installing : httpd-tools-2.4.6-95.el7.centos.x86_64 3/5
Installing : mailcap-2.1.41-2.el7.noarch 4/5
Installing : httpd-2.4.6-95.el7.centos.x86_64 5/5
Verifying : httpd-tools-2.4.6-95.el7.centos.x86_64 1/5
Verifying : mailcap-2.1.41-2.el7.noarch 2/5
Verifying : apr-1.4.8-7.el7.x86_64 3/5
Verifying : httpd-2.4.6-95.el7.centos.x86_64 4/5
Verifying : apr-util-1.5.2-6.el7.x86_64 5/5
Installed:
httpd.x86_64 0:2.4.6-95.el7.centos
Dependency Installed:
apr.x86_64 0:1.4.8-7.el7 apr-util.x86_64 0:1.5.2-6.el7 httpd-tools.x86_64 0:2.4.6-95.el7.centos mailcap.noarch 0:2.1.41-2.el7
Complete!
[root@localhost network-scripts]#
```

ảnh: Yum thành công

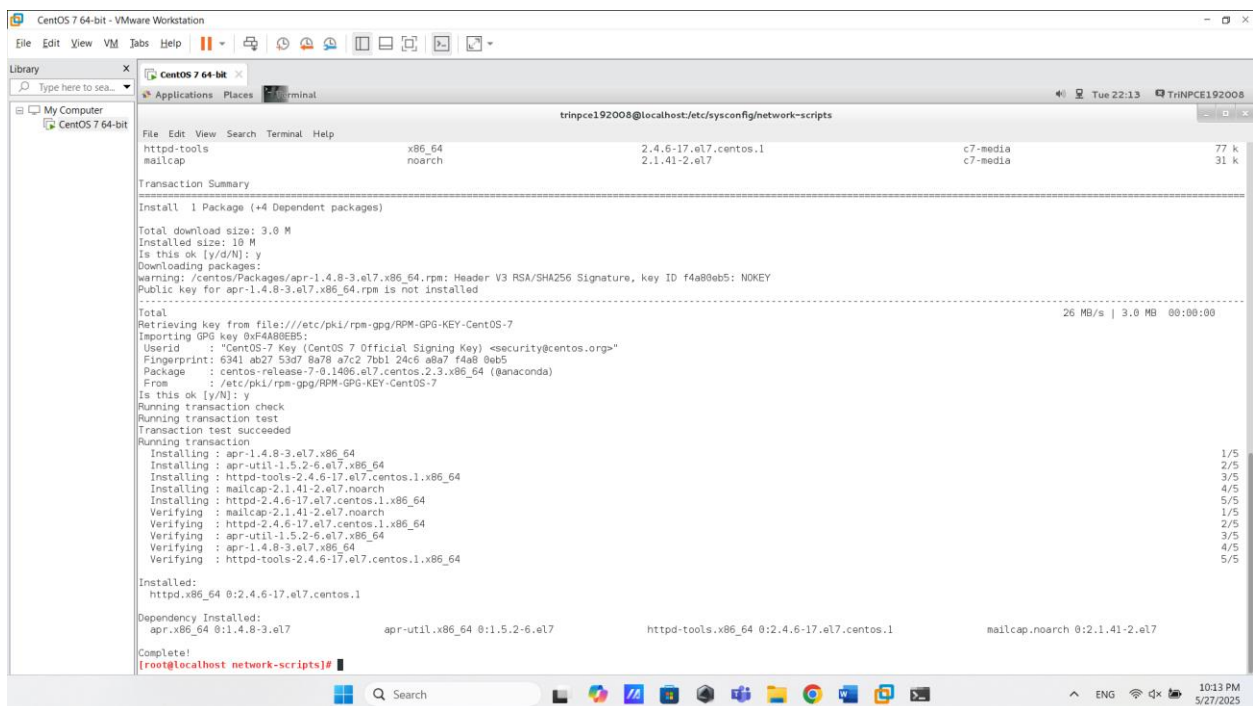
2. Yum từ CD



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
CentOS 7 64-bit
trinpce192008@localhost/etc/sysconfig/network-scripts
GNU nano 2.3.1 File: /etc/yum.repos.d/CentOS-Media.repo Modified
[c7-media]
name=CentOS-$releasever - Media
baseurl=file:///media/CentOS/
file:///media/cdrom/
file:///media/cdrecorder/
file:///centos/
gpgcheck=1
enabled=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7
```



2 ảnh trên: Mount CD, Tạo file repo từ CD, Sử dụng YUM từ đĩa để cài gói



ảnh trên: Yum thành công

Iptable:

```
[root@localhost network-scripts]# systemctl stop firewalld
[root@localhost network-scripts]# systemctl disable firewalld
[root@localhost network-scripts]# systemctl mask firewalld
[root@localhost network-scripts]# yum install iptables-services
Loaded plugins: fastestmirror, langpacks
base                                     | 3.6 kB  00:00:00
extras                                 | 3.6 kB  00:00:00
updates                               | 3.6 kB  00:00:00
(1/8): c7-media/group_gz              | 157 kB  00:00:00
(2/8): c7-media/primary_db            | 2.7 MB  00:00:00
(3/8): base/group_gz                  | 153 kB  00:00:01
(4/8): extras/group_gz                | 153 kB  00:00:00
(5/8): extras/primary_db              | 6.1 MB  00:00:01
(6/8): updates/group_gz               | 153 kB  00:00:00
(7/8): base/primary_db                | 6.1 MB  00:00:04
(8/8): updates/primary_db             | 6.1 MB  00:00:01
Determining fastest mirrors
* c7-media:
Resolving Dependencies
--> Running transaction check
--> Package iptables-services.x86_64 0:1.4.21-35.el7 will be updated
--> Package iptables-services.x86_64 0:1.4.21-35.el7 will be an update
--> Processing Dependency: iptables = 1.4.21-35.el7 for package: iptables-services-1.4.21-35.el7.x86_64
--> Running transaction check
--> Package iptables.x86_64 0:1.4.21-35.el7 will be updated
--> Package iptables.x86_64 0:1.4.21-35.el7 will be an update
--> Finished Dependency Resolution

Dependencies Resolved

=====================================================================================================================================
Package                               Arch          Version         Repository      Size
=====================================================================================================================================
Updating:
iptables-services                    x86_64        1.4.21-35.el7   base            52 k
Updating for dependencies:
iptables                             x86_64        1.4.21-35.el7   base            432 k
Transaction Summary
Upgrade 1 Package (+1 Dependent package)
```

```
--> Finished Dependency Resolution

Dependencies Resolved

=====================================================================================================================================
Package                               Arch          Version         Repository      Size
=====================================================================================================================================
Updating:
iptables-services                    x86_64        1.4.21-35.el7   base            52 k
Updating for dependencies:
iptables                             x86_64        1.4.21-35.el7   base            432 k
Transaction Summary
Upgrade 1 Package (+1 Dependent package)

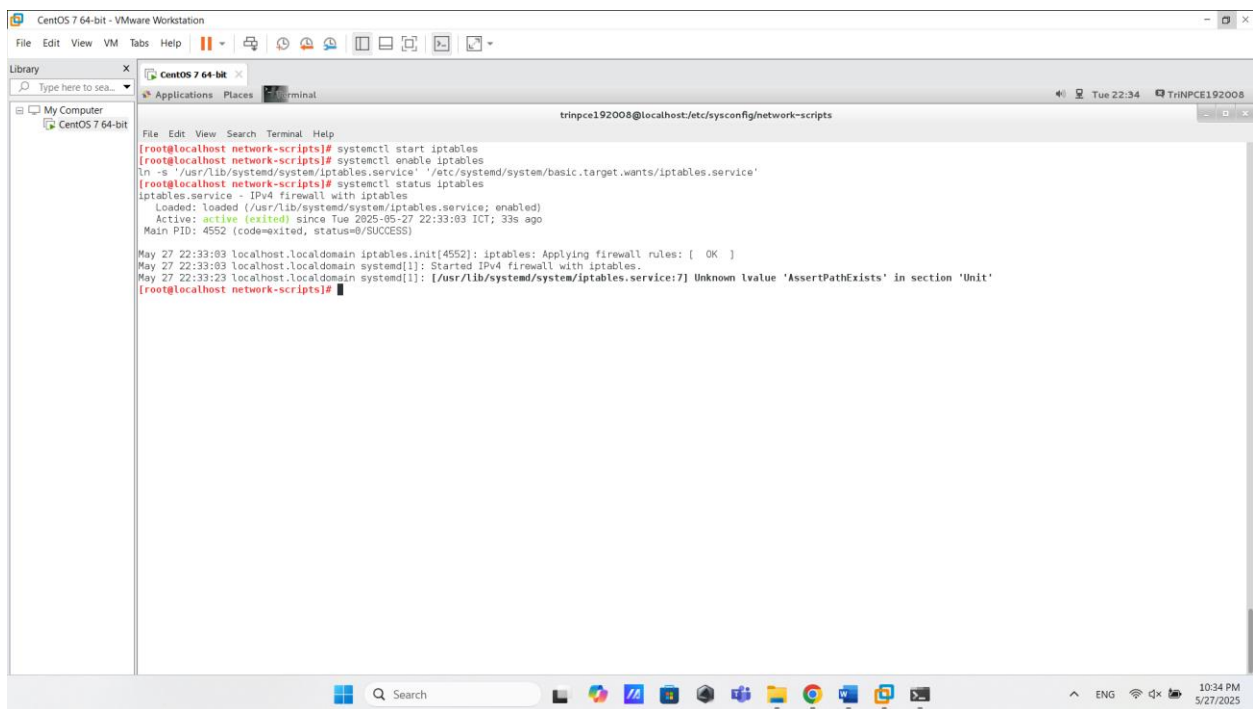
Total download size: 485 k
Is this ok [y/d/N]: y
Downloading packages:
No Presto metadata available for base
(1/2): iptables-services-1.4.21-35.el7.x86_64.rpm | 52 kB  00:00:01
(2/2): iptables-1.4.21-35.el7.x86_64.rpm         | 432 kB  00:00:02
-----
Total                                           289 kB/s | 485 kB  00:00:02
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Updating : iptables-1.4.21-35.el7.x86_64                                           1/4
  Updating : iptables-services-1.4.21-35.el7.x86_64                                2/4
  Cleanup  : iptables-services-1.4.21-13.el7.x86_64                                3/4
  Cleanup  : iptables-1.4.21-13.el7.x86_64                                           4/4
  Verifying : iptables-services-1.4.21-35.el7.x86_64                                1/4
  Verifying : iptables-1.4.21-35.el7.x86_64                                           2/4
  Verifying : iptables-services-1.4.21-13.el7.x86_64                                3/4
  Verifying : iptables-1.4.21-13.el7.x86_64                                           4/4

Updated:
iptables-services.x86_64 0:1.4.21-35.el7

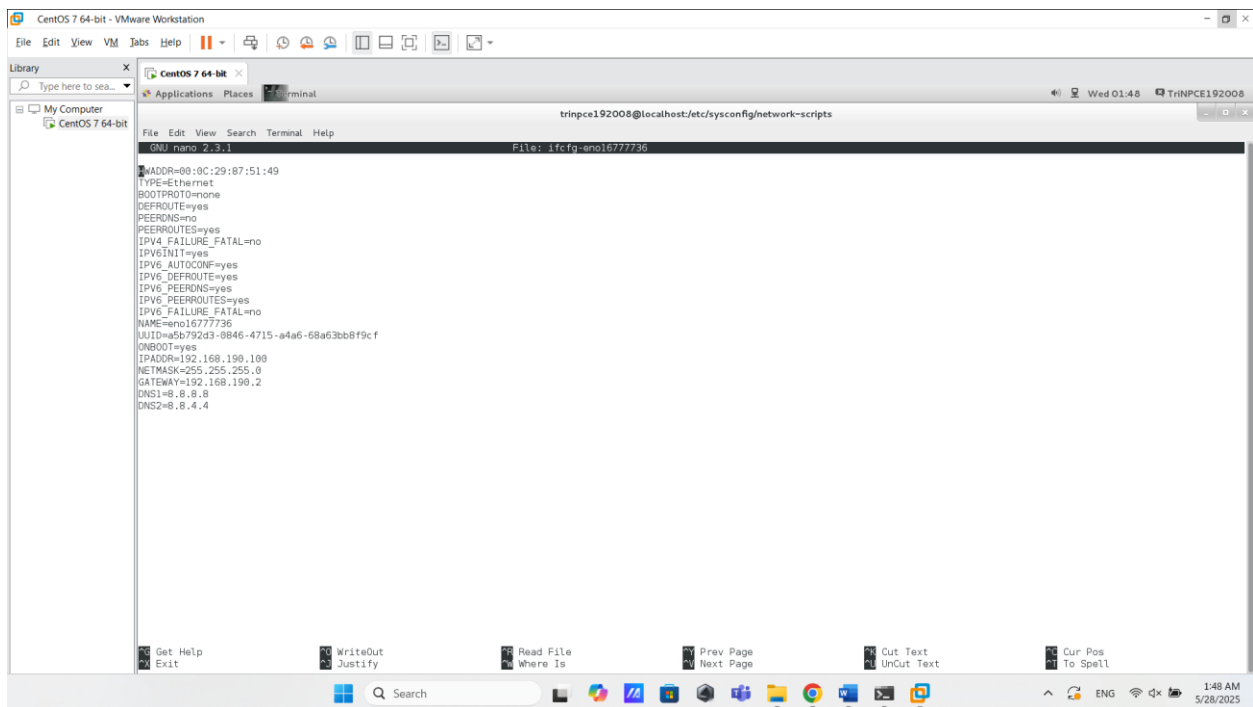
Dependency Updated:
iptables.x86_64 0:1.4.21-35.el7

Complete!
[root@localhost network-scripts]#
```

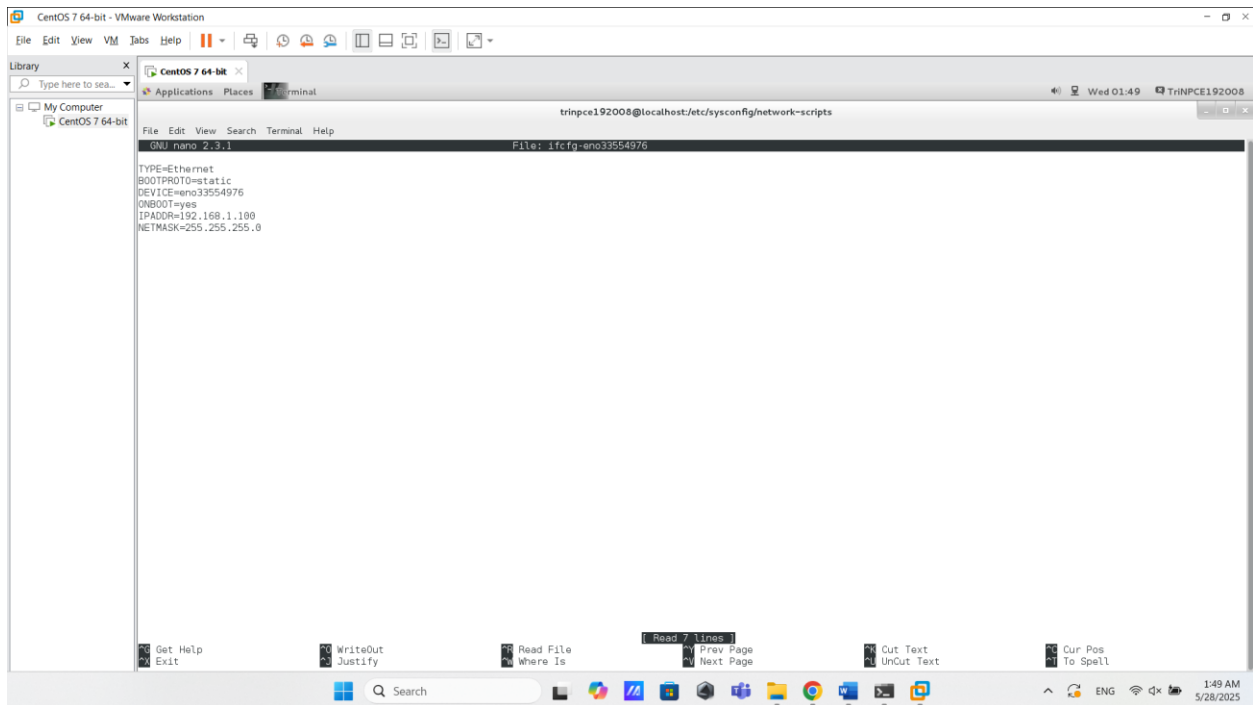
2 ảnh trên: tắt dịch vụ Firewalld, Không cho phép firewalld tự bật khi reboot server, Đảm bảo không cho các dịch vụ khác start firewalld, Cài đặt packages iptables-services từ CentOS repositories



ảnh trên: Khởi động dịch vụ IPTABLES, Bật tự động iptables khi boot server, để đảm bảo server luôn luôn có sự bảo vệ từ iptables, Kiểm tra trạng thái iptables đảm bảo nó đang hoạt động

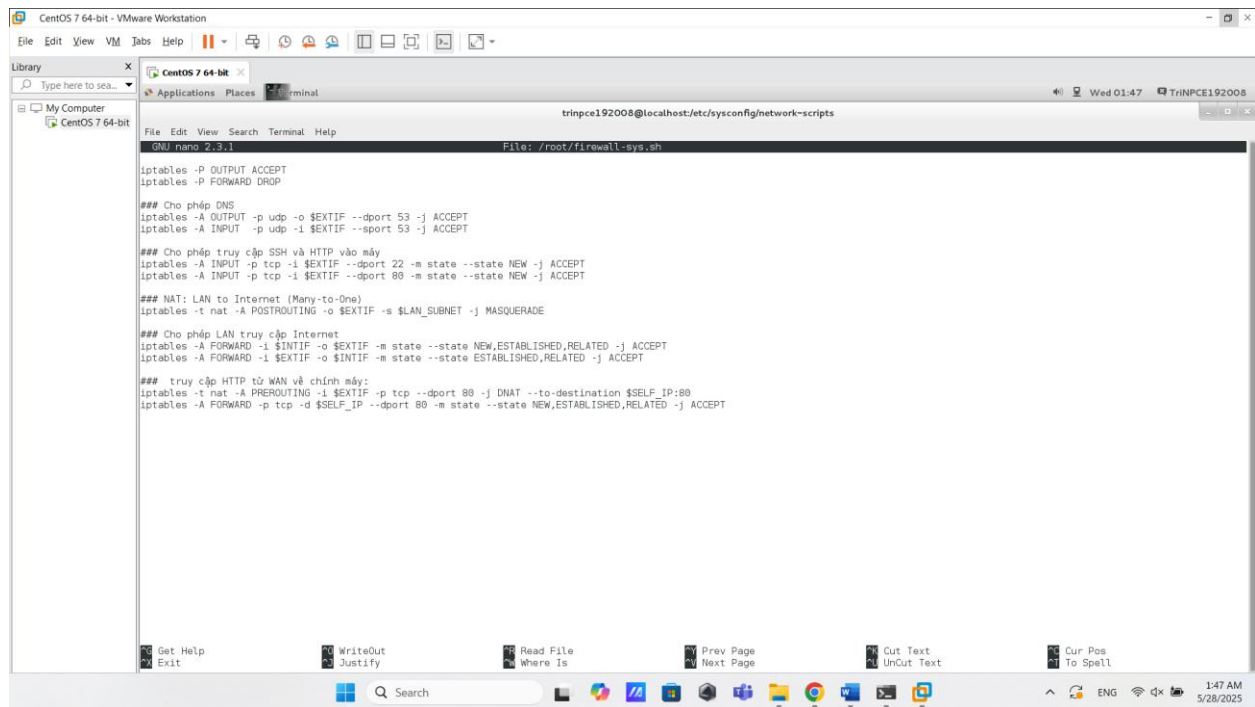


Ảnh trên: cấu hình card mạng WAN



ảnh trên: cấu hình card mạng LAN





2 ảnh trên: Tạo script iptables NAT + Port Forward:

```
#!/bin/bash
```

```
EXTIF="eno16777736"
```

```
INTIF="eno33554976"
```

```
LAN_SUBNET="192.168.1.0/24"
```

```
SELF_IP="192.168.1.100"
```

```
modprobe ip_tables
```

```
modprobe iptable_nat
```

```
modprobe ip_conntrack
```

```
modprobe ip_nat_ftp
```

```
echo 1 > /proc/sys/net/ipv4/ip_forward
```

Xóa luật cũ

iptables -F

iptables -t nat -F

iptables -X

Chính sách mặc định

iptables -P INPUT ACCEPT

iptables -P OUTPUT ACCEPT

iptables -P FORWARD DROP

Cho phép DNS

iptables -A OUTPUT -p udp -o \$EXTIF --dport 53 -j ACCEPT

iptables -A INPUT -p udp -i \$EXTIF --sport 53 -j ACCEPT

Cho phép truy cập SSH và HTTP vào máy

iptables -A INPUT -p tcp -i \$EXTIF --dport 22 -m state --state NEW -j ACCEPT

iptables -A INPUT -p tcp -i \$EXTIF --dport 80 -m state --state NEW -j ACCEPT

NAT: LAN to Internet (Many-to-One)

iptables -t nat -A POSTROUTING -o \$EXTIF -s \$LAN_SUBNET -j MASQUERADE

Cho phép LAN truy cập Internet

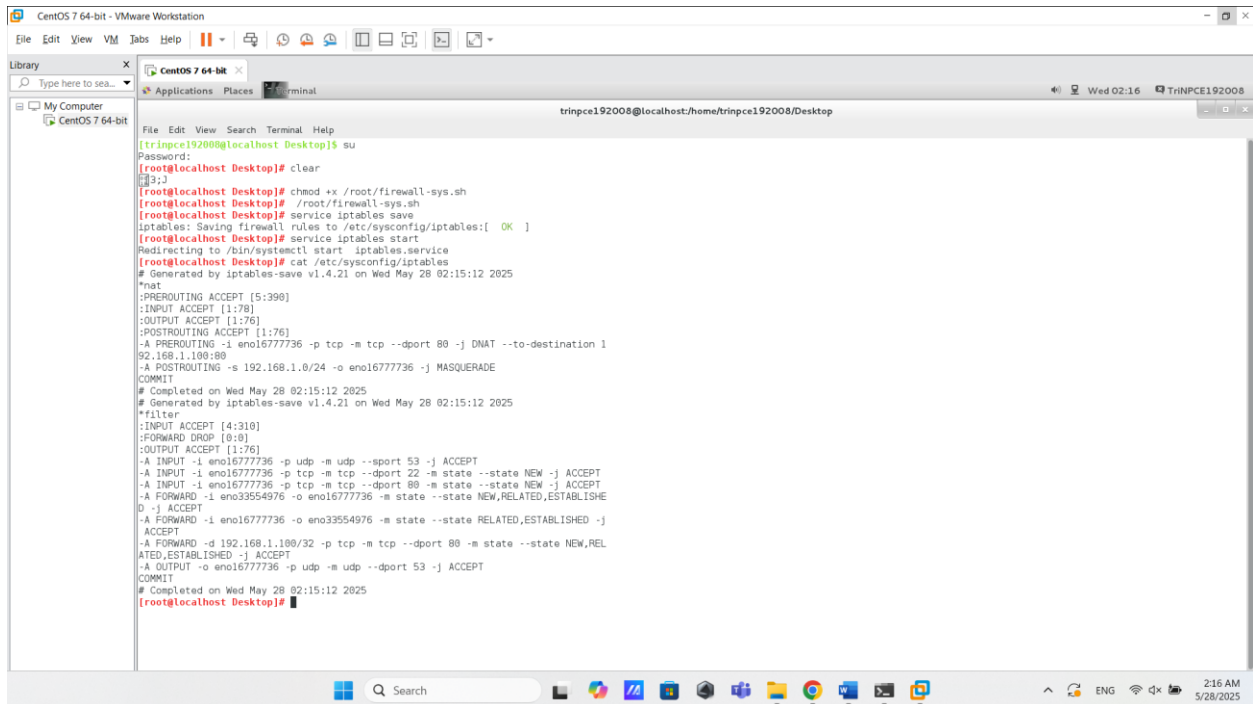
iptables -A FORWARD -i \$INTIF -o \$EXTIF -m state --state NEW,ESTABLISHED,RELATED -j ACCEPT

iptables -A FORWARD -i \$EXTIF -o \$INTIF -m state --state ESTABLISHED,RELATED -j ACCEPT

truy cập HTTP từ WAN về chính máy:

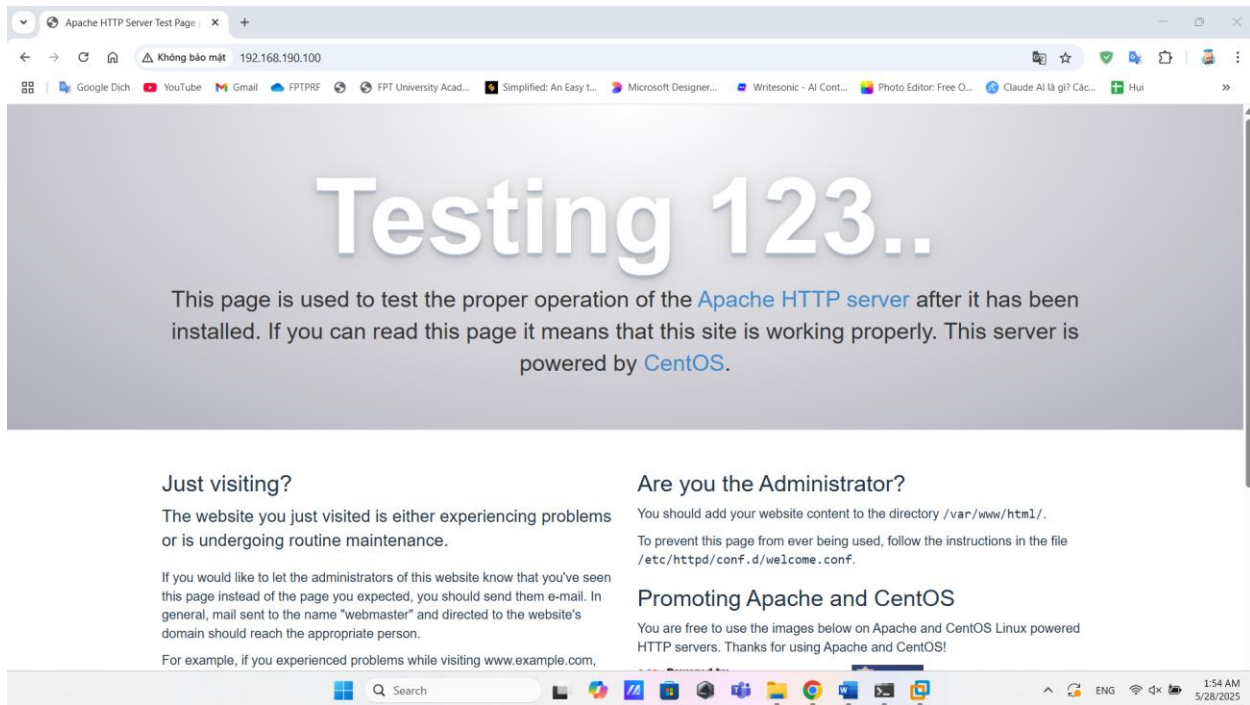
iptables -t nat -A PREROUTING -i \$EXTIF -p tcp --dport 80 -j DNAT --to-destination \$SELF_IP:80

iptables -A FORWARD -p tcp -d \$SELF_IP --dport 80 -m state --state NEW,ESTABLISHED,RELATED -j ACCEPT

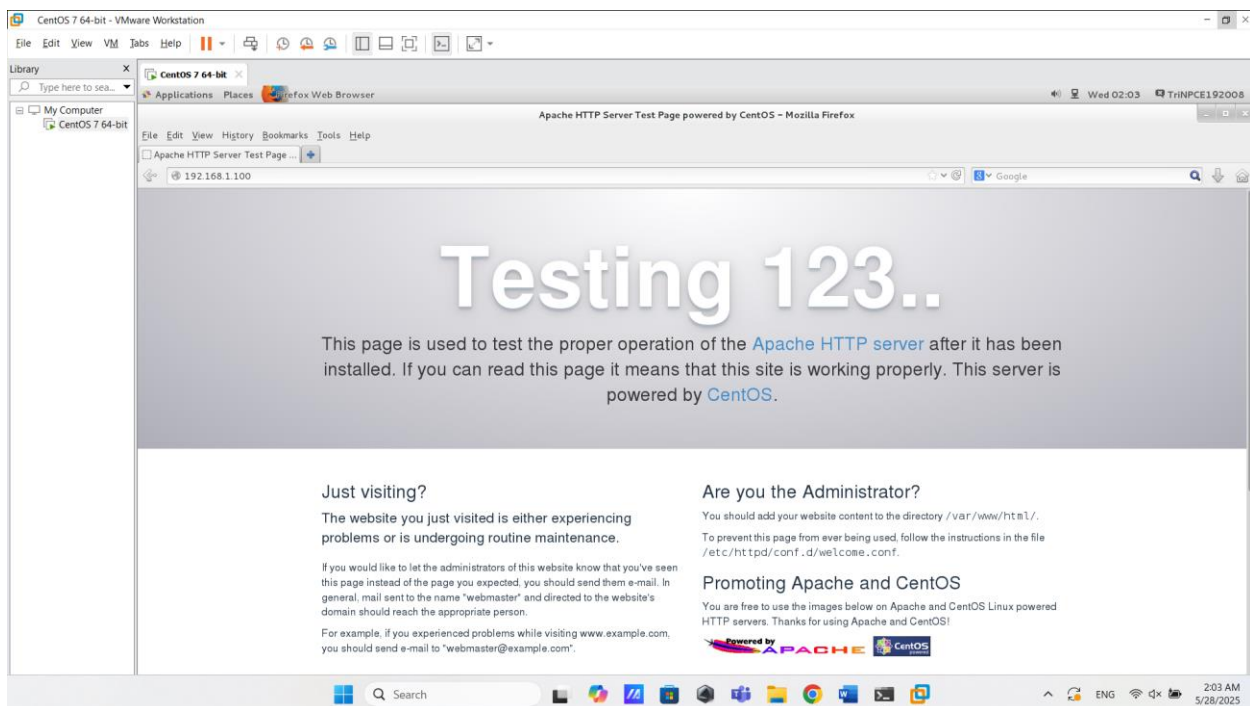


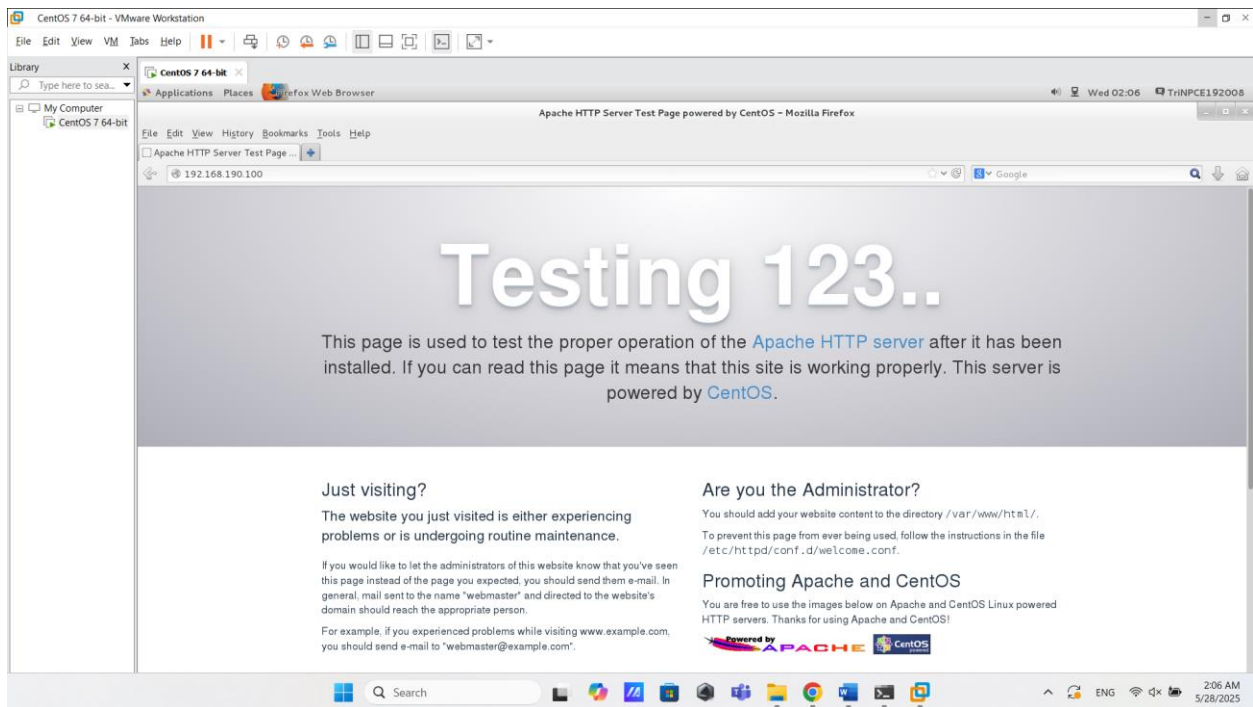
```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
CentOS 7 64-bit
My Computer
CentOS 7 64-bit
File Edit View Search Terminal Help
trinpce192008@localhost/home/trinpce192008/Desktop
[trinpce192008@localhost Desktop]$ su
Password:
[trinpce192008@localhost Desktop]$ clear
[trinpce192008@localhost Desktop]$
[trinpce192008@localhost Desktop]$ chmod +x /root/firewall-sys.sh
[trinpce192008@localhost Desktop]$ /root/firewall-sys.sh
[trinpce192008@localhost Desktop]$ service iptables save
iptables: Saving firewall rules to /etc/sysconfig/iptables:[ OK ]
[trinpce192008@localhost Desktop]$ service iptables start
Redirecting to /bin/systemctl start iptables.service
[trinpce192008@localhost Desktop]$ cat /etc/sysconfig/iptables
# Generated by iptables-save v1.4.21 on Wed May 28 02:15:12 2025
*nat
:PREROUTING ACCEPT [5:390]
:INPUT ACCEPT [1:78]
:OUTPUT ACCEPT [1:76]
:POSTROUTING ACCEPT [1:76]
-A PREROUTING -i eno16777736 -p tcp -m tcp --dport 80 -j DNAT --to-destination 1
92.168.1.100:80
-A POSTROUTING -s 192.168.1.0/24 -o eno16777736 -j MASQUERADE
COMMIT
# Completed on Wed May 28 02:15:12 2025
# Generated by iptables-save v1.4.21 on Wed May 28 02:15:12 2025
*filter
:INPUT ACCEPT [4:310]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [1:76]
-A INPUT -i eno16777736 -p udp -m udp --sport 53 -j ACCEPT
-A INPUT -i eno16777736 -p tcp -m tcp --dport 22 -m state --state NEW -j ACCEPT
-A INPUT -i eno16777736 -p tcp -m tcp --dport 80 -m state --state NEW -j ACCEPT
-A FORWARD -i eno33554976 -o eno16777736 -m state --state NEW,RELATED,ESTABLISHE
D -j ACCEPT
-A FORWARD -i eno16777736 -o eno33554976 -m state --state RELATED,ESTABLISHED -j
ACCEPT
-A FORWARD -d 192.168.1.100/32 -p tcp -m tcp --dport 80 -m state --state NEW,REL
ATED,ESTABLISHED -j ACCEPT
-A OUTPUT -o eno16777736 -p udp -m udp --dport 53 -j ACCEPT
COMMIT
# Completed on Wed May 28 02:15:12 2025
[trinpce192008@localhost Desktop]$
```

ảnh trên: chạy script

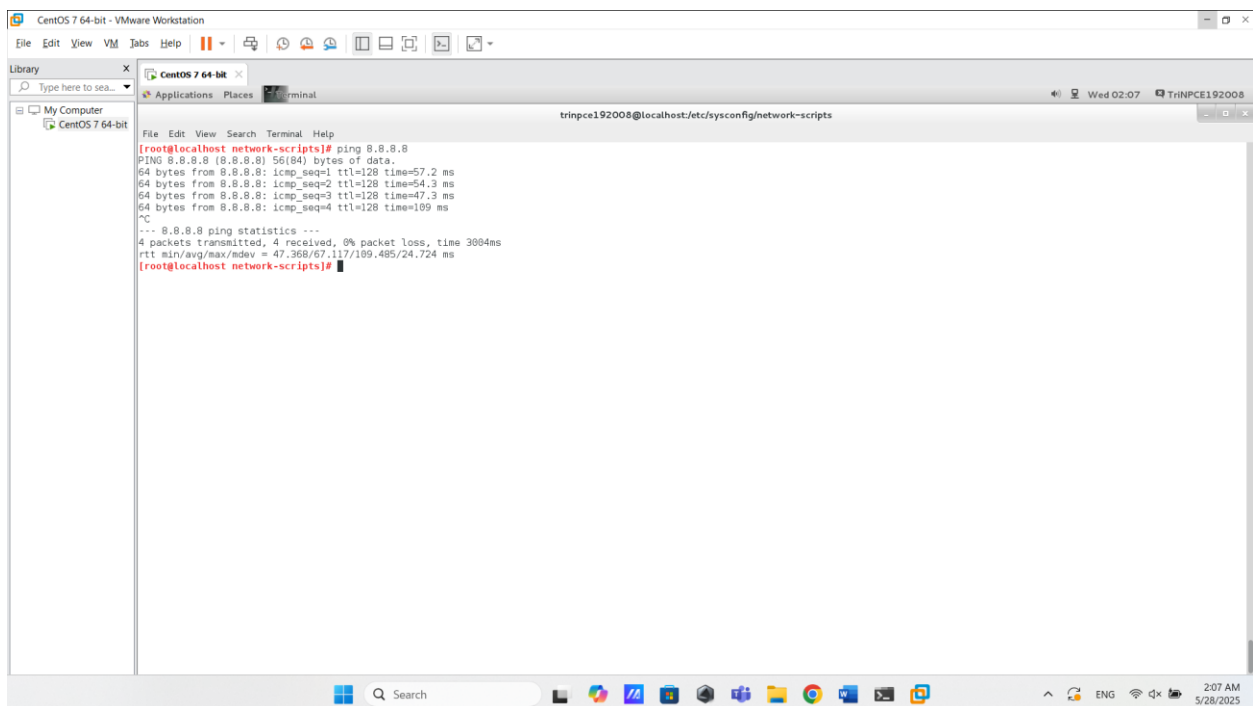


ảnh trên: Ping từ window

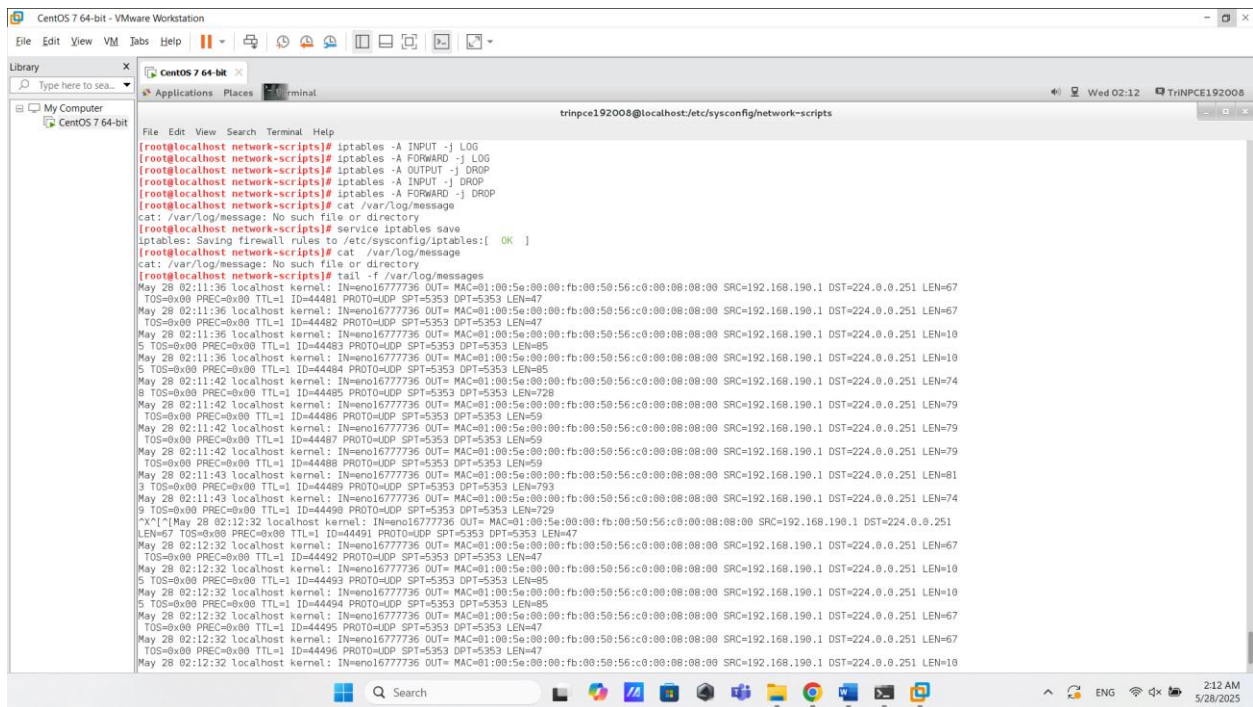




Ping từ web trên centos



ảnh trên: test DNS



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Help
Library
Type here to search
Applications Places
trinpce192008@localhost/etc/sysconfig/network-scripts

[root@localhost network-scripts]# iptables -A INPUT -j LOG
[root@localhost network-scripts]# iptables -A FORWARD -j LOG
[root@localhost network-scripts]# iptables -A OUTPUT -j DROP
[root@localhost network-scripts]# iptables -A INPUT -j DROP
[root@localhost network-scripts]# iptables -A FORWARD -j DROP
[root@localhost network-scripts]# cat /var/log/messages
cat: /var/log/messages: No such file or directory
[root@localhost network-scripts]# service iptables save
iptables: Saving firewall rules to /etc/sysconfig/iptables:[ OK ]
[root@localhost network-scripts]# cat /var/log/messages
cat: /var/log/messages: No such file or directory
[root@localhost network-scripts]# tail -f /var/log/messages
May 28 02:11:36 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=67
TOS=0x00 PREC=0x00 TTL=1 ID=44481 PROTO=UDP SPT=5353 DPT=5353 LEN=47
May 28 02:11:36 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=67
TOS=0x00 PREC=0x00 TTL=1 ID=44482 PROTO=UDP SPT=5353 DPT=5353 LEN=47
May 28 02:11:36 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=10
5 TOS=0x00 PREC=0x00 TTL=1 ID=44483 PROTO=UDP SPT=5353 DPT=5353 LEN=95
May 28 02:11:36 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=10
5 TOS=0x00 PREC=0x00 TTL=1 ID=44484 PROTO=UDP SPT=5353 DPT=5353 LEN=85
May 28 02:11:36 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=74
9 TOS=0x00 PREC=0x00 TTL=1 ID=44485 PROTO=UDP SPT=5353 DPT=5353 LEN=720
May 28 02:11:42 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=79
TOS=0x00 PREC=0x00 TTL=1 ID=44486 PROTO=UDP SPT=5353 DPT=5353 LEN=59
May 28 02:11:42 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=79
TOS=0x00 PREC=0x00 TTL=1 ID=44487 PROTO=UDP SPT=5353 DPT=5353 LEN=59
May 28 02:11:42 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=79
TOS=0x00 PREC=0x00 TTL=1 ID=44488 PROTO=UDP SPT=5353 DPT=5353 LEN=59
May 28 02:11:43 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=81
9 TOS=0x00 PREC=0x00 TTL=1 ID=44489 PROTO=UDP SPT=5353 DPT=5353 LEN=793
May 28 02:11:43 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=74
9 TOS=0x00 PREC=0x00 TTL=1 ID=44490 PROTO=UDP SPT=5353 DPT=5353 LEN=729
"X"["May 28 02:12:32 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251
LEN=67 TOS=0x00 PREC=0x00 TTL=1 ID=44491 PROTO=UDP SPT=5353 DPT=5353 LEN=47
May 28 02:12:32 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=67
TOS=0x00 PREC=0x00 TTL=1 ID=44492 PROTO=UDP SPT=5353 DPT=5353 LEN=47
May 28 02:12:32 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=10
5 TOS=0x00 PREC=0x00 TTL=1 ID=44493 PROTO=UDP SPT=5353 DPT=5353 LEN=85
May 28 02:12:32 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=10
5 TOS=0x00 PREC=0x00 TTL=1 ID=44494 PROTO=UDP SPT=5353 DPT=5353 LEN=85
May 28 02:12:32 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=67
TOS=0x00 PREC=0x00 TTL=1 ID=44495 PROTO=UDP SPT=5353 DPT=5353 LEN=47
May 28 02:12:32 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=67
TOS=0x00 PREC=0x00 TTL=1 ID=44496 PROTO=UDP SPT=5353 DPT=5353 LEN=47
May 28 02:12:32 localhost kernel: IN=eno1677736 OUT= MAC=01:00:5e:00:00:fb:00:50:56:c0:00:00:00:00 SRC=192.168.190.1 DST=224.0.0.251 LEN=10
5 TOS=0x00 PREC=0x00 TTL=1 ID=44497 PROTO=UDP SPT=5353 DPT=5353 LEN=85
```

ảnh trên: lưu cấu hình iptable và cấu hình để xem log